Ga Hing Woo (Jiaxing Hu)

Year 13 student, Mount Albert Grammar School · Firmware and embedded Linux · Auckland, New Zealand

gahing@gahingwoo.com huhuvmb88@gmail.com gahingwoo.com github.com/gahingwoo
linkedin.com/in/ga-hing-woo-906aa7423 orcid.org/0009-0002-0840-8951

Content as of 14 November 2026. Sources: gahingwoo.com/evidence · this page: gahingwoo.com/cv

I bring up new Rockchip silicon on mainline software and send the work upstream. Five of my commits are in the mainline kernel, merged in the 7.3 window and present since v7.3-rc1, applied by the IOMMU, Rockchip and networking maintainers. The first OP-TEE port for the RK3576 and a Trusted Firmware-A secure-boot fix are merged. I reverse-engineered the RK3576's NPU from vendor model files to a working driver stack, and when a published result of mine turned out to be wrong I retracted it by name. Kernel work is signed off as Jiaxing Hu; everywhere else I am gahingwoo.

Upstream contributions, accepted

Linux, iommu/rockchip: two fixes for the RK3576 NPU's IOMMU instances

Mainline, v7.3-rc1 · 2026

Take every clock the devicetree provides instead of a fixed pair, so register writes behind extra clock gates are no longer silently dropped; and acknowledge a stale page fault left by boot firmware before enabling stall, which otherwise hangs the poll. Applied by Will Deacon, IOMMU co-maintainer, with his own Signed-off-by.

`git.kernel.org: 841363ebb508` `github: 841363ebb508` `git.kernel.org: b10d5920cafa` `github: b10d5920cafa` `lore: Will Deacon, applied`

Linux, arm64 dts: ArmSoM CM5 and CM5-IO board support

Mainline, v7.3-rc1 · 2026

Devicetree and binding for a new RK3576 compute module and carrier, through five review rounds. Acked-by Krzysztof Kozlowski on the binding; applied by Heiko Stübner, Rockchip SoC maintainer.

`git.kernel.org: f60f44c24b10` `github: f60f44c24b10` `git.kernel.org: 44a48a23974b` `github: 44a48a23974b` `lore: Heiko Stübner, applied`

Linux, net/phy motorcomm: enable the reference clock for the YT8521

Mainline, v7.3-rc1 · 2026

Fix for crystal-less RGMII PHYs that need a SoC-sourced 25 MHz reference. Reviewed-by Andrew Lunn; Tested-by Gavin Gao on a different board; applied by Jakub Kicinski, net maintainer. An earlier attempt in the MAC driver was withdrawn on maintainer advice and redone at the PHY layer.

`git.kernel.org: dc3b7209b5d3` `github: dc3b7209b5d3` `lore: v1, Reviewed-by Andrew Lunn` `lore: v2, Tested-by Gavin Gao`

Armbian: board maintainer for the ArmSoM CM5-IO

merged 27 Oct 2026

Edge kernel and mainline U-Boot for the CM5-IO, merged by Igor Pečovnik, with me named as the board's maintainer. On 2 November the RK3576 NPU series went into Armbian's edge kernel for the CM5-IO and the ROCK 4D. A code owner for the board and the Rockchip kernel configs and patches, and a member of the Armbian GitHub organisation. Support for a second RK3576 board, the LCKFB TaishanPi 3M, is in review since 10 November.

`github: armbian/build#11097` `github: armsom-cm5-io.csc` `github: armbian/build#11162` `github: armbian/build#11324`

OP-TEE: first RK3576 platform port

merged · 2026

Ported the open-source Trusted Execution Environment to a new Rockchip SoC. Passed all 57 automated checks, reproduced on separate hardware, Reviewed-by from a Cherry Embedded Solutions maintainer. Flipper Devices later ran it on Flipper One: 114 xtest cases, 35,681 subtests, zero failures, credited by name in their developer documentation.

`github: optee_os#7821` `github: platform_rk3576.c on master` `docs.flipper.net/one/dev-log/3`

Trusted Firmware-A: removed a redundant secure-boot override

merged · 2026

A copy-paste build-configuration defect that would have silently broken secure boot on any build without a secure OS. Reviewed by engineers at STMicroelectronics, Arm and Rockchip.

`TF-A change 51089`

Under review

Linux: RK3576 NPU enablement, 16 patches, PATCH v16

posted 8 Nov 2026

Bindings, power domains, IOMMU and accel/rocket changes to run the RK3576's NPU on the mainline driver. Reviewed-by ×10 (Heiko Stübner ×6, Abel Vesa ×2, Krzysztof Kozlowski, Igor Paunovic), Acked-by ×2 (Conor Dooley), Tested-by ×2 on RK3588 hardware I do not own. Tomeu Vizoso, the driver's author, replied to v15 and v16 answers him; Ulf Hansson has offered to take the power-domain patches. The root cause of the long-standing failure was a register field one word wide: a 16-bit task count where the RK3588 header assumed 12, confirmed on the list by Rockchip.

`lore: PATCH v16 00/16` `lore: Tomeu Vizoso on v10` `lore: Rockchip confirms the field layout`

Linux, pmdomain/rockchip: restore only the QoS that was saved

posted 1 Nov 2026

On the RK3576 the GPU's QoS registers read all zeros after its first power-on, on plain mainline. Found by measurement on a CM5-IO; Reviewed-by Shawn Lin and Finley Xiao of Rockchip.

`lore: the patch`

Mesa: RK3576 support in the Teflon delegate, !43804

posted · 2026

Five patches for one regular convolution on the rocket/Teflon path, deliberately the bare minimum the maintainer asked for first. Not yet reviewed.

`gitlab.freedesktop.org: mesa!43804`

Linux, media: RFC V4L2 driver for the RK3576 VEPU510 H.264 encoder

RFC · 2026

From-scratch encoder driver. Reviewed by Heiko Stübner and Nicolas Dufresne; on their advice, abandoned the stateful V4L2 interface in favour of the DRM/Vulkan-Video split being built for the RK3588.

`gitlab.collabora.com: RK3576 mainline-status, VEPU510 reference point`

What is next

RK3576 NPU series: into the accel tree. v16 is posted and answers Tomeu Vizoso and Heiko Stübner on v15; Ulf Hansson has offered to take the power-domain patches, and Armbian's edge kernel already carries v14. The goal is for the 16 patches to be applied, so the NPU works on a stock kernel.

Mesa: after the single regular convolution in !43804, the depthwise, pointwise and image-input layer types MobileNet is built from, as separate series.

charsiu: an open runtime that matches or beats the vendor's performance on the mainline driver, with no vendor code in the path. Nine models run end to end, Llama-3.2-1B with output identical to the exact CPU arithmetic. Against the vendor runtime on the same board, Llama-3.2-1B decodes 1.39x faster at 594 MHz and prompts are ahead to 602 tokens; against Rockchip's published table, time to first token reached parity on all four models on 18 September, on figures not yet written up.

VEPU510 encoder: rebase on the DRM/Vulkan-Video split being built for the RK3588 once it lands, as the maintainers advised.

edk2-rk3576: a complete, polished UEFI for the RK3576. It boots a mainline Fedora 44 desktop on the CM5-IO today, and Windows 11 on the CM5-IO and the TaishanPi 3M; the goal is every board it builds for verified and the platform support upstreamable.

Publication

RK3576 NPU bring-up on the open driver stack: preprint, version 2

Zenodo · 2026

Version 1 reported that chained compute failed below what software could reach. Version 2 retracts that on its first page: the cause was a register field-layout error in software, and MobileNet V1 now runs end to end on the NPU and returns the CPU's class. Most of version 2 is about why a disprove-first method held a false conclusion for 38 days. Covered by CNX Software; five register-model findings independently cross-confirmed on RK3588 by a separate reverse-engineering effort.

`doi:10.5281/zenodo.21990992 (v2)` `doi:10.5281/zenodo.21348017 (v1, withdrawn result)`

`github: gregordinary/rockchip-npu-notes, SOURCES.md`

Selected projects

charsiu: an LLM runtime that talks to the mainline NPU driver directly, no Mesa and no vendor runtime. Nine models run end to end at four bits, Llama-3.2-1B with output identical to the exact CPU arithmetic, plus a vision tower, Whisper and CLIP on the same primitives. C. github.com/gahingwoo/charsiu

edk2-rk3576: a TianoCore UEFI port for the RK3576 that boots a mainline Fedora 44 desktop on the CM5-IO, and Windows 11 on the CM5-IO and the LCKFB TaishanPi 3M. Builds for three boards and has run on all three; pinned on the Radxa forum by a Radxa engineer. C. github.com/gahingwoo/edk2-rk3576, forum.radxa.com

wddm-mali-g52: a Windows driver for the RK3576 Mali-G52, with Mesa's Panfrost as the user-mode half and my own kernel driver. On 13 November the Windows 11 desktop was composed on the Mali on one board, at a frame or two a second; a day later, with the CPU copies measured and removed, at up to 50. Direct3D 10_0, one display mode. C. github.com/gahingwoo/wddm-mali-g52

kiln: local LLM and vision inference on the RK3576 NPU against a mainline kernel, using the vendor driver out of tree as a stopgap. Subject of a step-by-step tutorial in Radxa's official documentation. github.com/gahingwoo/kiln, docs.radxa.com

rp2350-tz-tee: a minimal TrustZone-M Secure/Non-Secure example on the RP2350, with the SecureFault verified over SWD rather than trusted from source. C. github.com/gahingwoo/rp2350-tz-tee

SoC-Consistency: static analysis of devicetree power trees, rule packs for five chip families, runs in CI. On PyPI; packaged into Arch Linux by an unaffiliated maintainer. Python. github.com/gahingwoo/SoC-Consistency

RKDevelopTool-GUI: a Qt front end for Rockchip flashing on macOS and Linux. In the Arch AUR and on its own page in Radxa's documentation. Python, PySide6. github.com/gahingwoo/RKDevelopTool-GUI, docs.radxa.com

lore-enhancer: a browser extension that turns lore.kernel.org's raw patch pages into a mail-client view. JavaScript. github.com/gahingwoo/lore-enhancer

Review and community

Reviewed-by given twice on another contributor's accel/rocket patches, the second only after finding a real out-of-bounds write on v1 and seeing it fixed in v2. lore: clocks by name, lore: lifecycle fix v2

Armbian: approved a 6.1 zswap backport as a code owner, after checking it against linux-6.1.y and Rockchip's rk-6.1 and narrowing its stated reach to the zsmalloc case. github: armbian/build#11171

Collabora published its own summary of the NPU work; the Rockchip kernel maintainer commented on it. Collabora's RK3576 enablement tracker cites the NPU series and the encoder RFC as its reference points. linkedin.com/posts/collabora

EzySpeech: real-time transcription and translation into 21 languages, built for and still in use at Mt Albert Baptist Church's Sunday services, with a signed reference letter from the translation team leader. github.com/gahingwoo/ezy_speech_translate, mabc.org.nz/sundays, reference letter (PDF)

Mt Albert Baptist website: a rebuild of the church's site, live on a subdomain the church pointed at my GitHub Pages account; not yet their default site. new.mabc.org.nz, github.com/gahingwoo/mabc-ws

half-baked-studios: a GitHub organisation co-run with three other students, where I guide the group through embedded and firmware work. github.com/half-baked-studios

Education

Mount Albert Grammar School, Auckland

Year 13 · 2026

Secondary school student. The upstream, research and tooling work above was done alongside school.

Skills, as used in the work above

Linux kernel:
IOMMU, devicetree and bindings, pmdomain, PHY drivers, accel subsystem, V4L2; the mailing-list workflow with b4 and checkpatch, from RFC to applied.

Hardware bring-up:
Register-level reverse engineering, ordered register tracing and differential testing against a vendor stack, SWD debugging, power and clock trees.

Trusted execution and boot:
Trusted Firmware-A, OP-TEE, TrustZone-M on Cortex-M33, TianoCore EDK2 and UEFI.

Languages and tools:
C, Python, Shell, JavaScript; Qt/PySide6; Mesa and Teflon; GitHub Actions and CI.